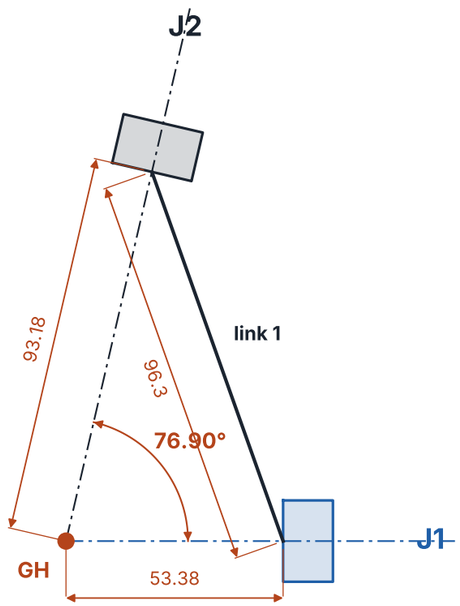
# 3R SHOULDER EXO - JOINT LAYOUT

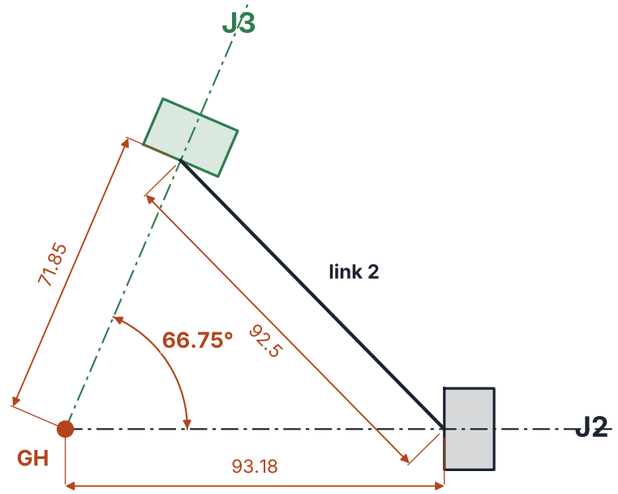
Right shoulder. Origin = GH centre (centre of the humeral head); every joint axis passes through it. Frame: x anterior, y up, z lateral. Units mm / deg. Pose: HOME = arm hanging straight down, all joints at 0.

LINK 1 - TRUE VIEW of the plane through GH containing the J1 and J2 axes



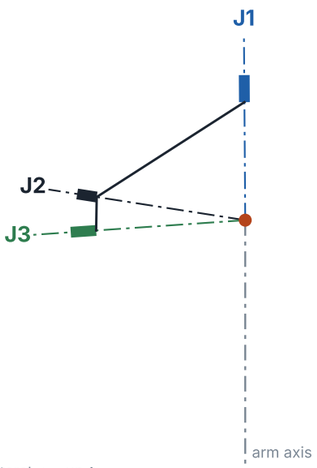

Dims: hub skin-side face distance from GH along its axis;
chord = hub face centre to hub face centre. Hub O20 x 12.25 stack.

LINK 2 - TRUE VIEW of the plane through GH containing the J2 and J3 axes



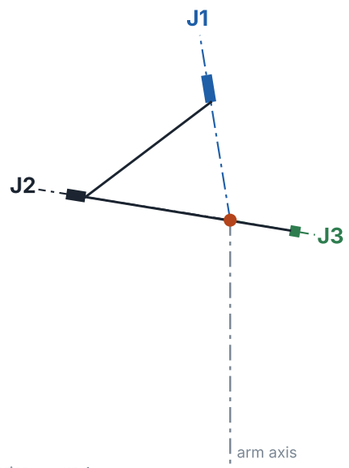

J2 is shared: link 1 and link 2 both pivot on it
(same axis, same 93.18 face).

FRONT (from in front)



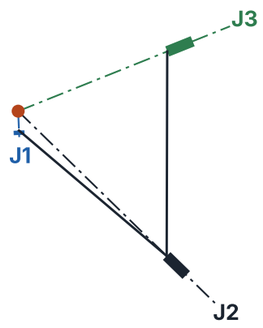

lateral <- up ^

SIDE (from the right side)




anterior -> up ^

TOP (from above)




lateral -> anterior ^

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JOINT AXES (unit vectors at HOME; all through GH)
           x       y       z  fixed to
J1   -0.1619 +0.9868 +0.0069  scapula
J2   -0.6937 +0.1109 +0.7117  links 1+2
J3   +0.3711 -0.0641 +0.9264  arm

HUB SKIN-SIDE FACE (distance from GH; hub
O20, stack 12.25 outward from the face)
  J1:  53.38 mm   centre (  -8.6,  +52.7,   +0.4)
  J2:  93.18 mm   centre ( -64.6,  +10.3,  +66.3)
  J3:  71.85 mm   centre ( +26.7,   -4.6,  +66.6)

ANGLES BETWEEN AXES
  J1-J2   76.90  (link 1, fixed)
  J2-J3   66.75  (link 2, fixed)
  J1-J3   96.71  (at HOME; changes with pose)

SEPARATION, face centre to face centre
  J1-J2    96.3 mm  (link 1, fixed)
  J2-J3    92.5 mm  (link 2, fixed)
  J1-J3    94.4 mm  (at HOME)

JOINT ROTATION over the 8 motion clips
  J1:  -46.9 ..  +63.0  (span 109.9)
  J2:  -36.9 ..  +82.5  (span 119.5)
  J3:  -33.4 ..  +84.8  (span 118.2)

J3 ON YOUR BOTTOM PART (arm-fixed)
  s = up the bar from the elbow-hinge
  point, across = bar width, normal = out
  of the bar face
  GH:      s  324.2  across  +0.0  normal  -78.9
  J3 face: s  324.3  across -26.7  normal  -12.2
  J3 axis to bar axis     89.9 deg
  J3 axis to bar normal   21.8 deg
  BOTTOM cut at s 255.0; kept bar 276.2 mm
  (eye end s -21.2 to the cut)

J1 ON THE SCAPULA HANDLER: its vector above
is scapula-fixed (scapula frame = this frame
at HOME).
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From the kinematics, cross-checked against the built CAD frames (design param2_r20_n3_s1414_4), 2026-10-15.